

Background

Mudskippers are an amphibious fish which exhibit a unique water-hopping behavior^[1]. When landing on the water’s surface, they create a water cavity that may transfer additional energy to subsequent jumps. This behavior has been observed with novel techniques^[2] and project aims to make the first mudskipper-inspired robotic model that can mimic water hopping.

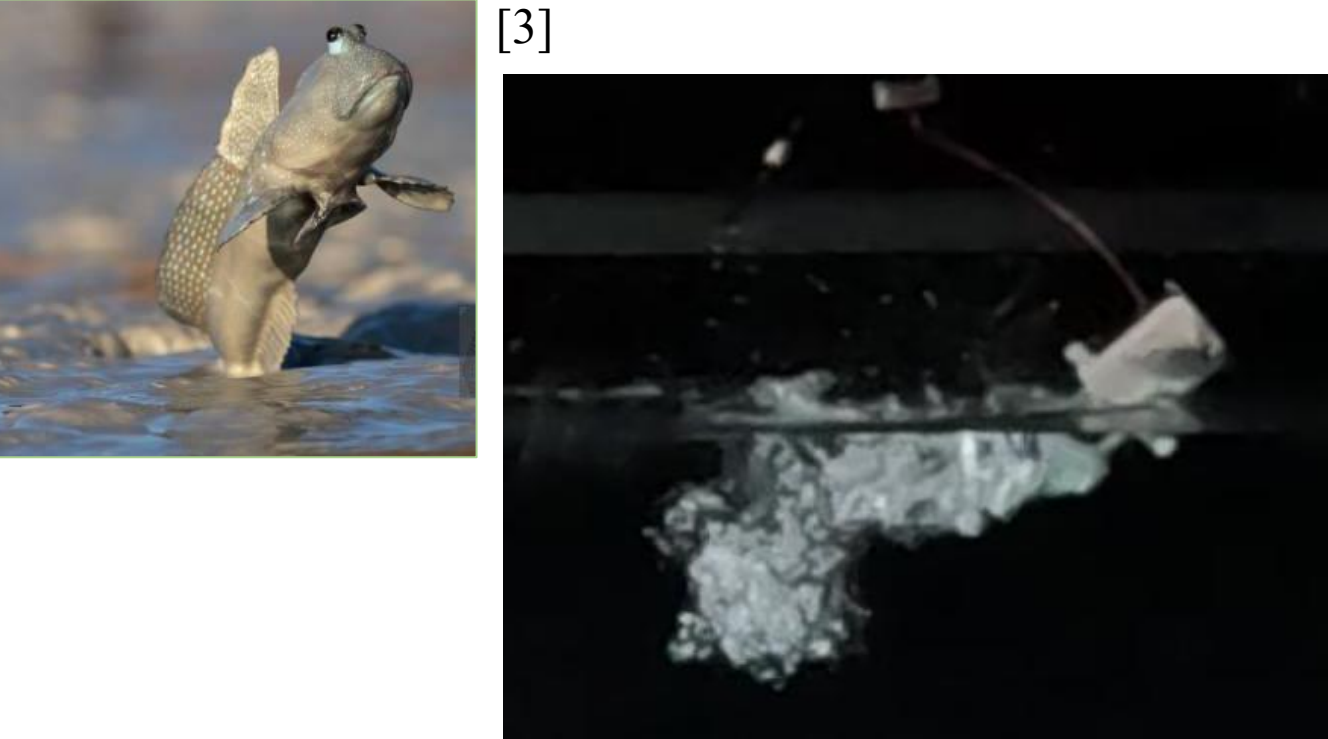


Fig. 1 Mudskipper fish water hopping (left), mudskipper robot hopping (right)

Objectives

Understand efficiency differences between water hopping and underwater swimming: **Does water hopping require less energy overall to move the same distance? How much of the efficiency difference comes from reduced drag in air and/or water cavity collapse?**

Understand how mudskipper water hopping mechanics work: **Can hopping behavior be accurately modeled with robots for further study?**

References

[1] Harris, V. A. (n.d.). On the locomotion of the mud-skipper *periophthalmus koelreuteri* (pallas): (gobiidae). *Proceedings of the Zoological Society of London*, 134(1), 107–135. <https://doi.org/10.1111/j.1469-7998.1960.tb05921.x>

[2] Choi, D., Yung, K., Bergerson, I., Wallace, H., Grafe, U., & Bhamla, S. (2025). *Three-dimensional tracking method for water-hopping mudskippers in natural habitats*. Cold Spring Harbor Laboratory. <https://doi.org/10.1101/2025.04.02.646728>

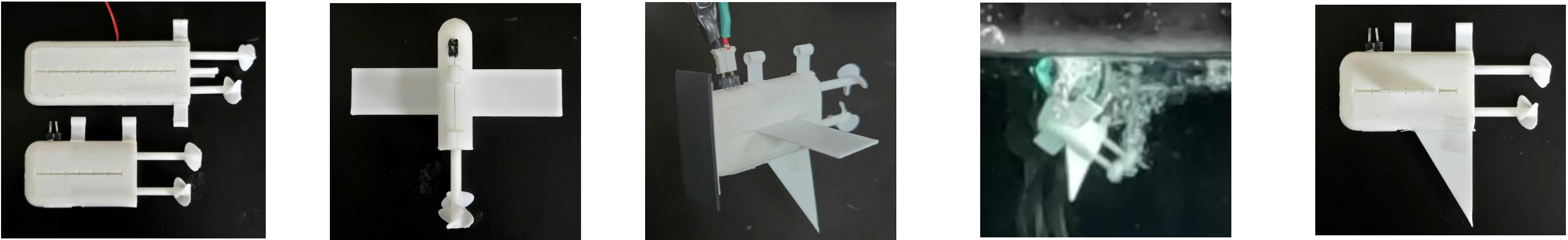
[3] Portelli, Scott (2024). *Blue spotted mudskipper jumping on the mudflats of Broome at low tide* [Photograph]. Getty Images. <https://www.gettyimages.com/detail/photo/jumping-mudskipper-royalty-free-image/2155809253>

Parameter Optimization

To accurately replicate hopping, voltage, robot fin location, fin size, and motor activation/deactivation timing were varied.

Parameter	Voltage	Fin Location	Fin Size	Duty Cycle	Caudal Fin
Reason for Varying	Counteract gravity and downward momentum	Optimize vertical lift and reduce pitching	Balance slamming force and drag	Thrust generation, hop rhythm and repeatability	Decrease rolling, increase # of hops

Fig. 2 From left to right: new robot model (bottom), fin location and size relative to body, drag enhancer, motor activation in water cavity, caudal fin



Stability and Roll Tests

Roll tests were conducted using a high-speed camera and a Hero 12 GoPro (240 fps) to capture the robot’s initial hop. Caudal and dihedral fins were added later to determine whether stability could be improved. Tracking and balancing the robot’s center of mass may also improve stability.

Fig. 3 Robot emerging from water surface during initial hop. Rolling behavior seen in the tilting of the robot body as it exits

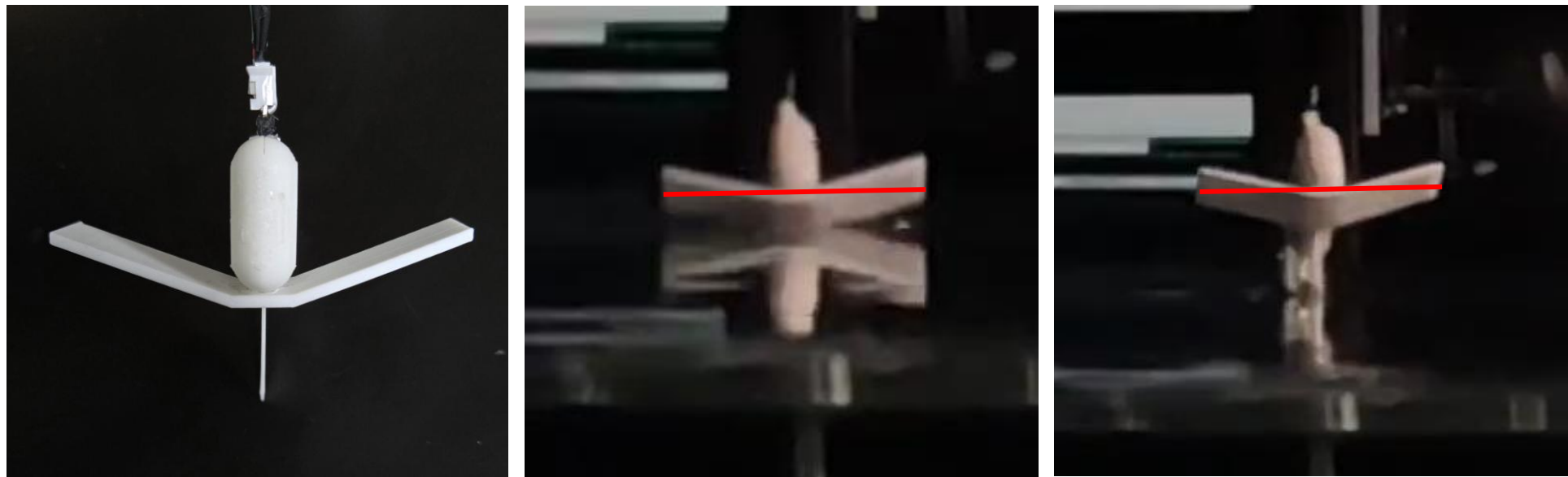
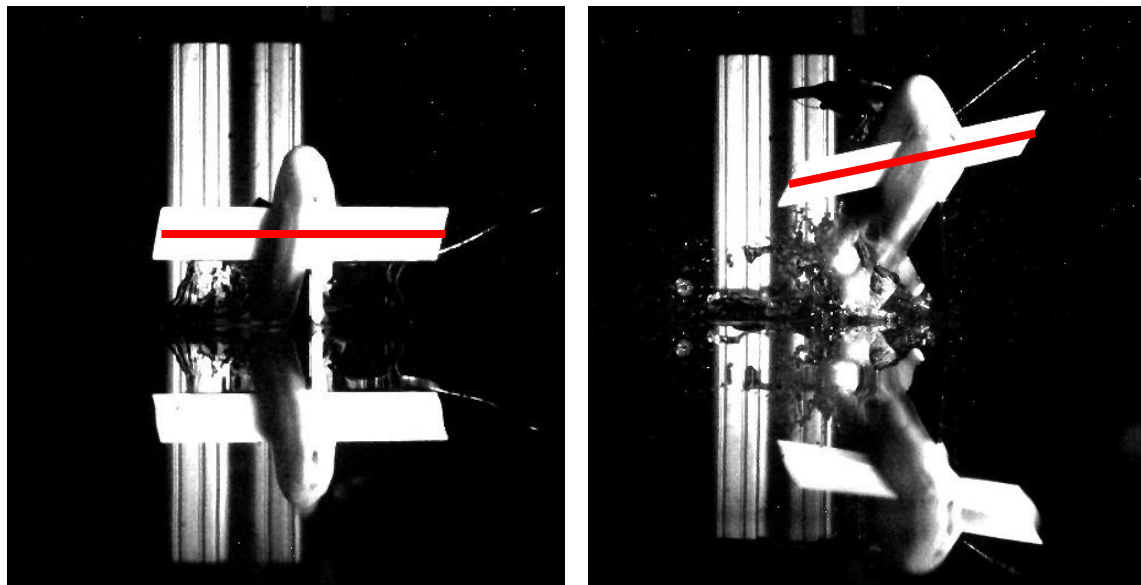


Fig. 4 (above) Addition of dihedral fin on robot (replacing original fins) reduces rolling behavior, seen in figure as less severe tilting when the robot exits the water

Hopping Efficiency

Five trials each were conducted of robot hopping and underwater swimming. Hopping was performed at 19V and swimming at 7.12V; the decrease in voltage is proportional to the duty cycle. Drag enhancers were used to simulate higher speeds but resulted in degradation of performance and reduction of efficiency. Position and velocities were found using DLTdv8 and MATLAB data processing.

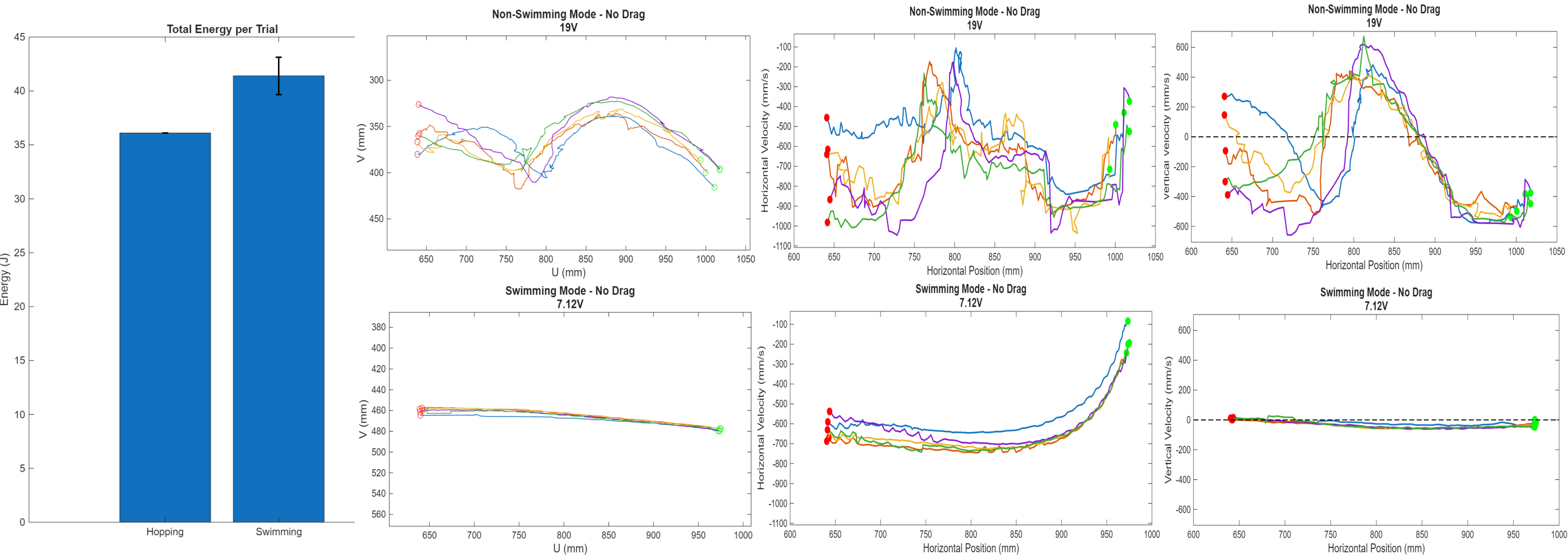


Fig. 5 Average total energy across 5 trials for hopping and swimming

Fig. 6 MATLAB plots of hopping (top) and swimming (bottom) for overall trajectory (left), horizontal velocity (middle) and vertical velocity (right)

Setup & Methods

All trials were conducted in an enclosed water tank and recorded using a GoPro at 240 fps. Robot begins fully submerged, positioned using a specifically designed bracket and steel wire. Motors activate in a three-pulse repeating pattern to create a jump. A guidewire made of fishing line was used to keep hops straight. An INA219 current sensor recorded current and voltage, from which power was calculated. Position was tracked from videos using DLTdv8 and data (position and power) was processed using MATLAB. Robot body design was simplified to reduce extra internal space; characteristic length is 40 mm.

Results

Parameter Optimization:

Voltage (V)	Fin Location (0 at left end of robot body, 1 at right end)	Fin Size (mm)	Duty Cycle (ms on/ms off)	Caudal Fin (mm)
19	0.5	15x30x0.6	70/115	30

Stability:

- Current model exhibits **rolling** – forward trajectory skews, reduces number of possible hops
- Reducing wire tension and dihedral fin addition improved stability – further improvements may **increase number of hops**

Efficiency:

- Hopping **reduces total travel energy** compared to swimming, mostly due to reduced drag in air. Some energy recapture from water cavity collapse may contribute
- Hopping behavior operates at:
 - $Re \approx 45,400$ and $We \approx 441$ based on water-surface impact velocity
 - $Fr \approx 0.88$ (from impact velocity) to 0.90 (from cycle-averaged velocity)

Next Steps

- Improve stability to increase number of hops to 3 or more
- Adjust fin shape and center of mass to assess impact on hopping
- Quantify energy transfer from cavity to subsequent hop